The algorithmlite package

A lightweight algorithm float without the float package

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https://github.com/munepi/algorithmlite

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1 Introduction

The algorithm package defines the `algorithm` float with `\newfloat` of the float package. The float package redefines `\caption` as a whole, for every float of the document, and it is often in the way of the class and of other packages that format captions.

The algorithmlite package defines the environments `algorithm` and `algorithm*` in the same way as the standard classes define `figure` and `table`: with `\@float` and `\@dblfloat` of the LaTeX kernel, so that the captions of algorithms are set by the `\@makecaption` of the class, as those of figures and tables. The float package is not loaded. If no other package gives the placement `[H]`, algorithmlite gives a minimal one.

The content of an algorithm is not the business of this package: load algorithmic, algpseudocode (algorithmicx) or any other package for it.

Algorithm 1: Euclid's algorithm

Require: $a \geq 0$, $b \geq 0$
1: **while** $b \neq 0$ **do**
2: $\quad (a, b) \leftarrow (b, a \bmod b)$
3: **end while**
4: **return** a

Supported formats/engines pdfLaTeX, XeLaTeX, LuaLaTeX (with or without LuaTeX-ja), pLaTeX and upLaTeX (with dvipdfmx). LaTeX2e 2020-10-01 or later is required.

2 Usage

```
\usepackage{algorithmlite}
\usepackage{algorithmic}   % or algpseudocode, ...

\begin{algorithm}[tbp]     % or [H]
\caption{Euclid's algorithm}\label{alg:euclid}
\begin{algorithmic}[1]
  ...
\end{algorithmic}
\end{algorithm}

\listofalgorithms
```

Algorithm 1 above is set with `[H]`. The package has no options.

3 What is defined

`algorithm`, `algorithm*` The float and its two-column variant, as `figure` and `figure*`. The default placement (`\fps@algorithm`) is `tbp`.

The counter `algorithm` With `\chapter`, it is reset at every chapter, and `\thealgorithm` is the chapter number, a period and ⟨*n*⟩ (just ⟨*n*⟩ before the first chapter), as `\thefigure` of the book classes; otherwise it is ⟨*n*⟩. Redefine `\thealgorithm` for another numbering.

`\algorithmname Algorithm`; in Japanese (the word "algorithm" in katakana) with pTEX, upTEX and LuaTEX-ja, unless the class option `english` of jsclasses or the LuaTEX-ja classes is given. LuaTEX-ja may also be loaded after algorithmlite.

`\listalgorithmname List of Algorithms`; in Japanese, "algorithm" followed by the word for "list" (as "List of Figures" is in jsclasses).

`\fnum@algorithm` `\algorithmname`⟨*sep*⟩`\thealgorithm`, where ⟨*sep*⟩ is taken from `\fnum@figure` of the class if it is `\figurename`⟨*sep*⟩`\thefigure`: `\nobreakspace` in the standard classes, `\nobreak` in jsclasses and the LuaTEX-ja classes, `~` in the pLATEX standard classes, nothing in jlreq. Otherwise ⟨*sep*⟩ is `\nobreak` for the Japanese name and `\nobreakspace` for the English one.

`\listofalgorithms` The list of algorithms, from the file `.loa`. It is `\listoffigures` of the class with `\listalgorithmname` and `.loa`, so that it looks like the list of figures (a `\chapter*` or a `\section*`, the running heads, ...). The entries are set with `\l@algorithm`, which is `\l@figure` of the class.

The float type The float type `\ftype@algorithm` is a bit, as are `\ftype@figure` = 1 and `\ftype@table` = 2. It is the next free bit, 4 in most documents. The counter of the float package (also used by floatrow and newfloat) and that of memoir are shared: if one of them is already there, its value is taken and the counter is moved on; if the float or floatrow package is loaded later (e.g. by minted), its counter is moved past 4. The newfloat package alone starts at 8.

4 The placement `[H]`

At `\begin{document}`, if neither the float package nor the floatrow package is loaded and `\@float@HH` is not defined, algorithmlite defines the placement `[H]`: the float is put in a box "here", in the text, with `\intextsep` above and below it, and `\caption` works in it as in a float. As with the float package, `[H]` is then available for every float type: `figure`, `table`, `algorithm`, ...

In the box, `\@captype` is the float type and `\@fps` is `H`, as `\@xfloat` of the kernel sets them for the other placements: code that runs in the float, such as a hook in `\@floatboxreset` or the caption, can tell an `[H]` float from the others by `\@fps`. The `[H]` of the float package sets `\@captype` but not `\@fps`, which is then undefined (or has the value from outside the float).

If the float package is loaded (before or after algorithmlite), its `[H]` is used.

5 Errors

`Package 'algorithm' cannot be used with algorithmlite` The algorithm or algorithm2e package is loaded before or after algorithmlite. Both define the `algorithm` environment; load only one of them.

`The algorithm environment or counter is already defined` algorithmlite does not redefine them.

6 Limitations

- `algorithm*[H]` in two-column mode is not supported (as with the float package). In one-column mode it works.

- Vertical writing (tate) is outside the scope of this version.

7 Tests

`make test` compiles every test twice with pdfLaTeX, XeLaTeX, LuaLaTeX, pLaTeX and upLaTeX (+ dvipdfmx) and checks the log. `test-common.tex` checks the numbers, names, float types and `\ref` immediately, and the pages of `[H]` and `[p]` floats from records written to the `.aux` file at shipout (`ALGL-RESULT OK`).

Test	What it covers
`test-basic.tex`	numbering with and without chapters, names (English/Japanese), `\fnum@algorithm`, `[H]`, `[p]`, `figure[H]`, `algorithm*`, `\ref`, `\listofalgorithms`; article, book, jsclasses, LuaTeX-ja classes, jlreq, pLaTeX standard classes
`test-hyperref.tex`	`test-basic.tex` with hyperref: no duplicate destinations
`test-twocolumn.tex`	`algorithm*` and `[H]` in two-column mode
`test-compat.tex`	float before/after, newfloat, caption, memoir
`test-errors.tex`	the error messages

8 History

v1.0 (2026/10/26) First public release.

9 License

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